

```
In [7]: %%sql
SELECT TOP(5) * FROM Superstore_Data;
```

Running query in 'mssql+pyodbc://@DESKTOP-7EM0EMB/SuperStore?driver=ODBC+Driver+17+for+SQL+Server&trusted_connection=yes'

Out[7]:

Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_Name	Segment	Country	City	State	Postal_Co
1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42
2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42
3	CA-2016-138688	2016-06-12	2016-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90
4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33
5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33

EDA

1- Sales by City

2- Customer by Transactions

3- Sales by Segment

4- Customer Segmentation Based on Sales

5- Calculate Shipping Days for Each Order

6- Sales Over Time by Year

7- Top 10 Customers by Sales

8- Sales and Profit by Category

9- Which Cities generate high Sales but low Profit ?

10- Which Regions have the Highest Shipping Delays ?

Sales by City

In [111...

```
%%sql
SELECT TOP(10) City, CAST(SUM(Sales) AS DECIMAL(10,2)) AS Total_Sales,
CONCAT(CAST((SUM(Sales) * 100.0 / (SELECT SUM(Sales) FROM Superstore_Data)) AS DECIMAL(10,2)), '%')
AS Sales_Percentage
FROM Superstore_Data
GROUP BY City
ORDER BY Total_Sales DESC;
```

Running query in 'mssql+pyodbc://@DESKTOP-7EM0EMB/SuperStore?driver=ODBC+Driver+17+for+SQL+Server&trusted_connection=yes'

Out[111...

City	Total_Sales	Sales_Percentage
New York City	256368.16	11.16%
Los Angeles	175851.34	7.66%
Seattle	119540.74	5.20%
San Francisco	112669.09	4.90%
Philadelphia	109077.01	4.75%
Houston	64504.76	2.81%
Chicago	48539.54	2.11%
San Diego	47521.03	2.07%
Jacksonville	44713.18	1.95%
Springfield	43054.34	1.87%

In []:

Customer by Transactions

In [112...

```
%%sql
SELECT TOP(10) Customer_Name,
COUNT(Order_ID) AS Tran_num
FROM Superstore_Data GROUP BY Customer_Name ORDER BY Tran_num DESC;
```

Running query in 'mssql+pyodbc://@DESKTOP-7EM0EMB/SuperStore?driver=ODBC+Driver+17+for+SQL+Server&trusted_connection=yes'

Out[112...

Customer_Name	Tran_num
William Brown	37
John Lee	34
Matt Abelman	34
Paul Prost	34
Seth Vernon	32
Edward Hooks	32
Chloris Kastensmidt	32
Jonathan Doherty	32
Arthur Prichep	31
Zuschuss Carroll	31

In []:

Sales by Segment

In [113...

```
%%sql
SELECT Segment, CAST(SUM(Sales) AS DECIMAL(10,2)) Total_Sales
FROM Superstore_Data GROUP BY Segment ORDER BY Total_Sales DESC;
```

Running query in 'mssql+pyodbc://@DESKTOP-7EM0EMB/SuperStore?driver=ODBC+Driver+17+for+SQL+Server&trusted_connection=yes'

Out[113...

Segment	Total_Sales
Consumer	1161401.34
Corporate	706146.36
Home Office	429653.15

In []:

Customer Segmentation

```
In [120... %%sql
CREATE VIEW Customer_Levels AS
SELECT Customer_Name, CAST(SUM(Sales) AS DECIMAL(10,2)) AS Total_Sales,

CASE
    WHEN SUM(Sales) > 10000 THEN 'VIP'
    WHEN SUM(Sales) > 5000 THEN 'Average Customer'
    ELSE 'Normal Customer'
END AS Customer_Levels

FROM Superstore_Data
GROUP BY Customer_Name;
```

Running query in 'mssql+pyodbc://@DESKTOP-7EM0EMB/SuperStore?driver=ODBC+Driver+17+for+SQL+Server&trusted_connection=yes'

Out[120...

```
In [121... %%sql
SELECT Customer_Levels, COUNT(Customer_Levels) AS Customer_Count FROM Customer_Levels
GROUP BY Customer_Levels ORDER BY Customer_Count DESC;
```

Running query in 'mssql+pyodbc://@DESKTOP-7EM0EMB/SuperStore?driver=ODBC+Driver+17+for+SQL+Server&trusted_connection=yes'

```
Out[121... Customer_Levels Customer_Count
Normal Customer          676
Average Customer         97
VIP                      20
```

In []:

Calculate Shipping Days for Each Order

```
In [122... %%sql
SELECT TOP(10) Order_Date, Ship_Date,
```

```
DATEDIFF(DAY, Order_Date, Ship_Date) AS Shipping_Days FROM Superstore_Data;
```

Running query in 'mssql+pyodbc://@DESKTOP-7EM0EMB/SuperStore?driver=ODBC+Driver+17+for+SQL+Server&trusted_connection=yes'

Out[122...]

Order_Date	Ship_Date	Shipping_Days
2016-11-08	2016-11-11	3
2016-11-08	2016-11-11	3
2016-06-12	2016-06-16	4
2015-10-11	2015-10-18	7
2015-10-11	2015-10-18	7
2014-06-09	2014-06-14	5
2014-06-09	2014-06-14	5
2014-06-09	2014-06-14	5
2014-06-09	2014-06-14	5
2014-06-09	2014-06-14	5
2014-06-09	2014-06-14	5

In []:

Sales Over Time by Year

In [19]:

```
%%sql
SELECT YEAR(Order_Date) AS Order_Date, CAST(SUM(Sales) AS DECIMAL(10,2)) AS Total_Sales
FROM Superstore_Data
GROUP BY YEAR(Order_Date)
ORDER BY Order_Date;
```

Running query in 'mssql+pyodbc://@DESKTOP-7EM0EMB/SuperStore?driver=ODBC+Driver+17+for+SQL+Server&trusted_connection=yes'

Out[19]: **Order_Date** **Total_Sales**

2014	484247.50
2015	470532.51
2016	609205.60
2017	733215.25

In []:

Top 10 Customers by Sales

In [15]: `%%sql
SELECT TOP(10) Customer_Name, CAST(SUM(Sales) AS DECIMAL (10,2)) AS Total_Sales FROM
Superstore_Data GROUP BY Customer_Name ORDER BY Total_Sales DESC;`

Running query in 'mssql+pyodbc://@DESKTOP-7EM0EMB/SuperStore?driver=ODBC+Driver+17+for+SQL+Server&trusted_connection=yes'

Out[15]: **Customer_Name** **Total_Sales**

Sean Miller	25043.05
Tamara Chand	19052.22
Raymond Buch	15117.34
Tom Ashbrook	14595.62
Adrian Barton	14473.57
Ken Lonsdale	14175.23
Sanjit Chand	14142.33
Hunter Lopez	12873.30
Sanjit Engle	12209.44
Christopher Conant	12129.07

In [10]:

Sales and Profit by Category

In [23]:

```
%%sql
SELECT Category, CAST(SUM(Sales) AS DECIMAL(10,2)) AS Total_Sales, CAST(SUM(Profit) AS DECIMAL(10,2)) AS Total_Profit
FROM Superstore_Data GROUP BY Category ORDER BY Total_Sales DESC, Total_Profit DESC;
```

Running query in 'mssql+pyodbc://@DESKTOP-7EM0EMB/SuperStore?driver=ODBC+Driver+17+for+SQL+Server&trusted_connection=yes'

Out[23]:

Category	Total_Sales	Total_Profit
Technology	836154.03	145454.95
Furniture	741999.79	18871.27
Office Supplies	719047.03	122490.80

In []:

Which Cities generate high Sales but low Profit

In [26]:

```
%%sql
SELECT TOP(10) City, CAST(SUM(Sales) AS DECIMAL(10,2)) AS Total_Sales, CAST(SUM(Profit) AS DECIMAL(10,2)) AS Total_Profit
FROM Superstore_Data GROUP BY City HAVING SUM(Profit) < 1000
ORDER BY Total_Sales DESC;
```

Running query in 'mssql+pyodbc://@DESKTOP-7EM0EMB/SuperStore?driver=ODBC+Driver+17+for+SQL+Server&trusted_connection=yes'

Out[26]:

City	Total_Sales	Total_Profit
Philadelphia	109077.01	-13837.77
Houston	64504.76	-10153.55
Chicago	48539.54	-6654.57
Jacksonville	44713.18	-1903.83
San Antonio	21843.53	-7299.05
Burlington	21668.08	-3622.88
Dallas	20131.93	-2846.53
Charlotte	13693.34	-429.78
Louisville	12345.81	-1430.31
Denver	12198.79	-244.92

In []:

Which Regions have the Highest Shipping Delays

In [37]:

```
%%sql
SELECT Region, AVG(DATEDIFF(DAY, Order_Date, Ship_Date)) AS Avg_Shipping_Days
FROM Superstore_Data GROUP BY Region ORDER BY Avg_Shipping_Days DESC;
```

Running query in 'mssql+pyodbc://@DESKTOP-7EM0EMB/SuperStore?driver=ODBC+Driver+17+for+SQL+Server&trusted_connection=yes'

Out[37]:

Region	Avg_Shipping_Days
Central	4
East	3
South	3
West	3